| | | | | |
|---|---|---|---|---|
| **3** | **(a)** | | loss in gravitational potential energy = gain in kinetic energy + gain in elastic potential energy | B1 |
| | **(b)** | **(i)** | $4.0 \times 9.81 \times 0.090 = \frac{1}{2} \times 200 \times 0.09^2 + E_k$ | B1 |
| | | | $E_k = 2.7$ J | B1 |
| | | **(ii)** | Since both blocks have same acceleration and hence same velocity, $E_k \propto m$<br>So taking ratio, $\frac{E_{k,Q}}{E_k} = \frac{4}{6}$<br><br>or other appropriate method | M1 |
| | | | $E_k = 1.8$ J | A1 |
| | **(c)** | | When blocks come to rest $E_k = 0$. Let distance required by $x$, then conservation of energy gives<br>$$4 \times 9.81 \times x = \frac{1}{2} \times 200 \times x^2$$ | M1 |
| | | | $x = 39$ cm | A1 |

4 (a)